IIITB COMET FOUNDATION

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Gate -PH 2011 46 Question

QUESTION

The following Boolean expression

$$Y = A \bullet \overline{B} \bullet \overline{C} \bullet \overline{D} + \overline{A} \bullet B \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet C \bullet D + \overline{A} \bullet B \bullet C \bullet D + A \bullet \overline{B} \bullet \overline{C} \bullet D$$

can be simplified to

(A) $\overline{A} \bullet \overline{B} \bullet C + A \bullet \overline{D}$

(B) $\overline{A} \bullet B \bullet \overline{C} + A \bullet \overline{D}$

(C) $A \bullet \overline{B} \bullet \overline{C} + \overline{A} \bullet D$

(D) $A \bullet \overline{B} \bullet C + \overline{A} \bullet D$

QUESTION ANALYSIS

Solution

can be simplified to

(A) $\overline{A} \cdot \overline{B} \cdot C + A \cdot \overline{D}$ **(B)** $\overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{D}$

(C) $A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot D$ **(D)** $A \cdot \overline{B} \cdot C + \overline{A} \cdot D$

Solution

Given,

$$\begin{aligned} Y = {} & A\overline{BCD} + \overline{A}B\overline{C}D + \overline{ABC}D \\ & + \overline{AB}CD + \overline{A}BCD + A\overline{BC}D \end{aligned}$$

Step 1: Group the terms

Group the first and last terms:

$$A\overline{BCD} + A\overline{BC}D = A\overline{BC}(\overline{D} + D)$$

Since $\overline{D} + D = 1$,

$$A\overline{BC}(\overline{D} + D) = A\overline{BC}.$$

Now group the remaining four terms:

$$\overline{A}B\overline{C}D + \overline{ABC}D + \overline{AB}CD + \overline{A}BCD$$

$$= \overline{A}D\left(B\overline{C} + \overline{BC} + \overline{B}C + BC\right).$$

Step 2: Simplify inside the brackets

$$B\overline{C} + \overline{BC} + \overline{B}C + BC$$
$$= \overline{C}(B + \overline{B}) + C(B + \overline{B})$$
$$= \overline{C}(1) + C(1)$$
$$= \overline{C} + C$$
$$= 1.$$

Therefore,

$$\overline{A}D(1) = \overline{A}D.$$

Step 3: Final expression

Hence,

$$\boxed{Y = A\overline{BC} + \overline{A}D}$$

This matches **Option (C).**

$$\boxed{\text{Correct Answer: (C)}}$$

Hardware Components Required

S.No.	Component
1	ESP32-WROOM Development Board
2	Personal Computer (Ubuntu OS)
3	USB Cable
4	Breadboard
5	LED
6	220 Ω Resistor
7	Jumper Wires

Table 1: Hardware and Software Components Required

Hardware Connections

S.No.	ESP32-WROOM Pin	Connected To
1	GPIO 2	LED Anode (+)
2	LED Cathode (-)	220 Ω Resistor
3	220 Ω Resistor	GND
4	USB Port	Personal Computer (Ubuntu)

Table 2: Hardware Connections for LED Interface with ESP32-WROOM

Execution Procedure

1. Create a new PlatformIO project for the ESP32-WROOM board.

   ```
   /root/venv/bin/pio project init --board esp32dev
   ```

2. Place the source code (`main.cpp`) inside the `src` directory of the project.

3. Compile the source code and generate the binary files using:

   ```
   /root/venv/bin/pio run
   ```

4. After successful compilation, the following binary files are generated in the build directory:

   `.pio/build/esp32dev/`

   - `bootloader.bin`
   - `partitions.bin`
   - `firmware.bin`

5. Copy the generated binary files to the desired directory on the Ubuntu PC.

6. Connect the ESP32-WROOM board to the Ubuntu PC using a USB cable.

7. Verify that the ESP32 board is detected by the operating system.

   ```
   ls -l /dev/ttyUSB*
   ```

8. Verify the ESP32 chip.

   ```
   esptool.py --chip esp32 --port /dev/ttyUSB0 chip_id
   ```

9. Flash the generated binary files into the ESP32.

   ```
   esptool.py --chip esp32 \
     --port /dev/ttyUSB0 \
     --baud 460800 \
     write_flash \
     0x1000 bootloader.bin \
     0x8000 partitions.bin \
     0x10000 firmware.bin
   ```

10. Wait until the flashing process completes successfully.

11. Press the **EN (Reset)** button on the ESP32 board if required.

12. Observe the LED connected to the output GPIO pin to verify the implemented Boolean expression.

Results and Observations

The Boolean expression was successfully implemented on the ESP32-WROOM development board. The generated binary files were flashed onto the ESP32 using the `esptool.py` utility. The output was verified using an LED connected to the output GPIO pin. The LED turned ON for logic HIGH ($Y = 1$) and remained OFF for logic LOW ($Y = 0$), thereby confirming the correct implementation of the Boolean function.

Truth Table

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Table 3: Truth Table of the Implemented Boolean Expression

Experimental Output

Figure 1 shows the LED in the OFF state corresponding to logic LOW ($Y = 0$). Figure 2 shows the LED in the ON state corresponding to logic HIGH ($Y = 1$).

Result

The Boolean expression was successfully implemented and verified on the ESP32-WROOM development board. The observed LED output matched the expected truth table for all input combinations.

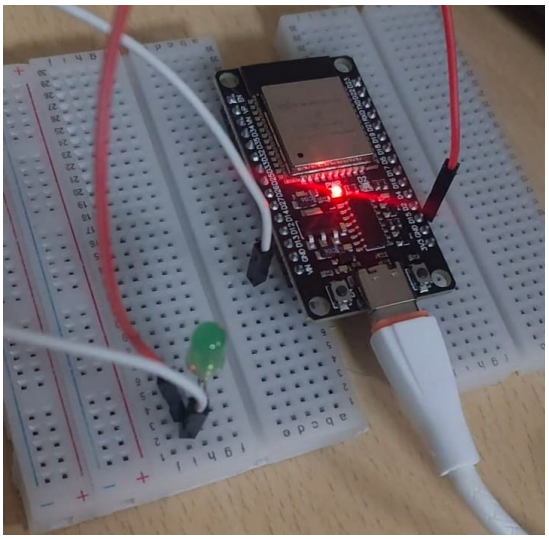


Figure 1: LED OFF (Logic 0)

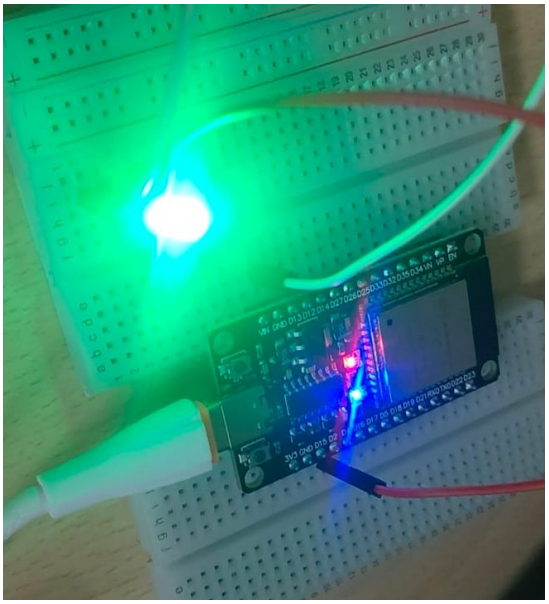


Figure 2: LED ON (Logic 1)